

- 1 An object of mass 350 g is released from rest at a height of 50 cm on a frictionless incline as shown in Fig. 1.1. The incline makes an angle of 30° to the horizontal. It is placed on a table of height 2.0 m above the smooth ground. The table is fixed to the ground.

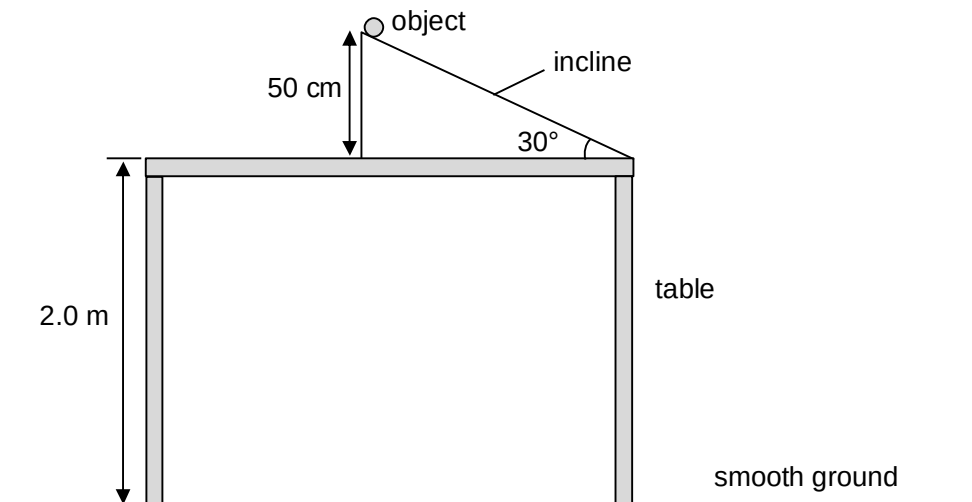


Fig. 1.1

- (a) Show that the acceleration of the object is 4.9 m s^{-2} as it slides down the incline.

[1]

- (b) Determine the speed of the object as it leaves the incline.

speed = m s^{-1} [3]

- (c) The object lands on the ground after falling off the incline.

Calculate the horizontal distance of the object from the edge of the table.

distance = m [3]

- (d) A small trolley of mass 1.2 kg is moving to the left at a speed of 4.0 m s^{-1} along the smooth ground. The object is released again from the same height on the incline. It lands in the trolley and they move off together horizontally.

- (i) Explain whether momentum of the system of object and trolley is conserved in the horizontal direction.

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 [2]

- (ii) Hence, calculate the final velocity of the trolley and object.

magnitude of velocity = m s^{-1}

direction = [3]

2. (a) The pressure p of an ideal gas of density ρ is related to the mean square speed $\langle c^2 \rangle$